AS MATHEMATICS - PURE 1

Quadratics from zero

Day 1: completing the square

Complete self-learning notes, guided practice and answers

YOUR AIM TODAY

You will understand what a quadratic is, learn why completing the square works, use it to find turning points, and solve simple quadratic equations.

Block	What you do	Time
1	Learn the language and repair the algebra basics	15 min
2	Copy and explain the worked examples	20 min
3	Complete the guided practice	20 min
4	Complete the independent practice	20 min
5	Attempt the exit check and self-mark	10 min

YOUR RULE

Keep pages 11-13 closed until you finish pages 1-10. Show every line of algebra. If an answer is wrong, correct the line where the mistake began - do not only copy the final answer.

WHAT YOU NEED

Pencil, eraser, ruler and about 85 focused minutes. No calculator is needed today.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

FOUNDATION 1

What is a quadratic?

Start with the words. They make the method easier to remember.

An **expression** is a collection of numbers, letters and operations. An **equation** says that two expressions are equal. A **quadratic** has highest power 2.

Word	Plain meaning	Example
Term	A part separated by + or - signs.	In $x^2 + 6x + 5$, the terms are x^2, $6x$ and 5.
Coefficient	A number multiplying a letter.	The coefficient of x in 6x is 6.
Constant	A number without x.	The constant in $x^2 + 6x + 5$ is 5.
Root / solution	An x-value that makes an equation true.	$x = 2$ is a root of $x^2 - 4 = 0$.

The general quadratic

$$ax^2 + bx + c, \text{ where } a \text{ is not zero}$$

- a is the coefficient of x^2.
- b is the coefficient of x.
- c is the constant.

EXAMPLE

For $2x^2 - 7x + 3$: **a = 2**, **b = -7** and **c = 3**. The negative sign belongs to 7.

Stop and answer aloud

- Is $4x + 1$ quadratic? Why not?
- For $x^2 - 8x - 9$, state a, b and c.
- What is the difference between an expression and an equation?

CHECK

Answers: No, its highest power is 1. For the second expression, a = 1, b = -8, c = -9. An equation contains an equals sign.

FOUNDATION 2

The algebra move behind the method

Completing the square depends on expanding a bracket correctly.

Multiply every term in the first bracket by every term in the second bracket.

$$(x + 3)(x + 3)$$

Multiply	Result
x by x	x^2
x by 3	3x
3 by x	3x
3 by 3	9

Add the middle terms: $x^2 + 3x + 3x + 9 = x^2 + 6x + 9$.

TWO PATTERNS

$(x + p)^2 = x^2 + 2px + p^2$
$(x - p)^2 = x^2 - 2px + p^2$

Mini-practice

Question	Your expansion
1. $(x + 2)^2$	
2. $(x + 5)^2$	
3. $(x - 4)^2$	
4. $(x - 7)^2$	

SELF-CHECK

Answers: 1. $x^2 + 4x + 4$ 2. $x^2 + 10x + 25$
3. $x^2 - 8x + 16$ 4. $x^2 - 14x + 49$

If any middle term is wrong, remember: the middle coefficient is **twice** the number inside the bracket.

CORE METHOD

What does completing the square mean?

Turn x squared plus bx into one squared bracket plus or minus a number.

We want to rewrite a quadratic in this form:

$$(x + p)^2 + q$$

The word **complete** means we create the exact constant needed to make a perfect square, then compensate so the value of the expression does not change.

THE RECIPE WHEN THE COEFFICIENT OF X SQUARED IS 1

1. Take the coefficient of x.
2. Divide it by 2. Put that number inside the bracket.
3. Square that number.
4. Subtract it outside the bracket.
5. Add the original constant.

Worked example 1

Write $x^2 + 6x + 1$ in completed-square form.

Step	Working	Reason
1	Half of 6 is 3.	The bracket is $(x + 3)^2$.
2	$(x + 3)^2 = x^2 + 6x + 9$.	The square creates an extra +9.
3	$x^2 + 6x + 1 = (x + 3)^2 - 9 + 1$	Subtract 9 to keep the expression unchanged.
4	$= (x + 3)^2 - 8$	Combine the constants.

FAST CHECK

Expand $(x + 3)^2 - 8$. You get $x^2 + 6x + 9 - 8 = x^2 + 6x + 1$, so the answer is correct.

Say the rule aloud: halve b, square it, subtract it, then add c.

Positive and negative x terms

The sign inside the bracket follows the sign of the x term.

Example 2: complete the square for $x^2 - 8x + 3$.

Move	Working
Half of -8	-4, so begin with $(x - 4)^2$.
Square -4	$(-4)^2 = 16$.
Compensate	$x^2 - 8x + 3 = (x - 4)^2 - 16 + 3$.
Simplify	**$(x - 4)^2 - 13$**.

SIGN WARNING

Half of -8 is -4. But $(-4)^2$ is +16. Squaring a negative number gives a positive result.

Example 3: complete the square for $x^2 + 5x + 1$.

Half of 5 is 5/2. Fractions are allowed.

$$x^2 + 5x + 1 = (x + 5/2)^2 - 25/4 + 1$$
$$= (x + 5/2)^2 - 25/4 + 4/4$$
$$= (x + 5/2)^2 - 21/4$$

DO NOT PANIC ABOUT FRACTIONS

The same recipe works. Keep fractions exact: 1 = 4/4, so -25/4 + 1 = -21/4.

Before moving on

- Explain why the outside number in Example 2 is -13.
- Expand each final answer to check that it returns to the original expression.

MEANING

Turning points and minimum values

Completed-square form tells you important graph facts immediately.

A square is never negative: $(\text{anything})^2$ is always at least 0.

KEY FACT

If $y = (x - h)^2 + k$, the turning point is **(h, k)**. The minimum value of y is k because the squared part can be 0 but cannot be negative.

Example 4

$$y = x^2 - 8x + 3 = (x - 4)^2 - 13$$

Question	Answer	Why
When is the square zero?	When $x - 4 = 0$, so $x = 4$.	Then $(x - 4)^2 = 0$.
Turning point	(4, -13)	Use (h, k).
Minimum y-value	-13	The square cannot go below 0.
Range	y is greater than or equal to -13	Every output is -13 or higher.
Line of symmetry	$x = 4$	The graph mirrors around the turning point.

BRACKET SIGN TRAP

For $(x - 4)^2$, the turning-point x-coordinate is +4 because $x - 4 = 0$ at $x = 4$. For $(x + 3)^2$, it is -3.

Quick check

For $y = (x + 2)^2 - 7$, state the turning point, minimum value, line of symmetry and range.

Your answer:

Check: turning point (-2, -7); minimum -7; line $x = -2$; range y is greater than or equal to -7.

Solve an equation by completing the square

Make the squared bracket the subject, then take both square roots.

Solve $x^2 + 6x + 1 = 0$.

From page 4: $x^2 + 6x + 1 = (x + 3)^2 - 8$.

Line	Working	What happened
1	$(x + 3)^2 - 8 = 0$	Use completed-square form.
2	$(x + 3)^2 = 8$	Add 8 to both sides.
3	x + 3 = +sqrt(8) or -sqrt(8)	A positive square has two square roots.
4	x = -3 + sqrt(8) or x = -3 - sqrt(8)	Subtract 3.
5	x = -3 + 2sqrt(2) or x = -3 - 2sqrt(2)	Simplify sqrt(8) = sqrt(4 x 2).

MOST COMMON MISTAKE

Do not write only the positive square root. If $A^2 = 8$, then A = +sqrt(8) **or** A = -sqrt(8).

A simpler example

Solve $x^2 - 4x - 5 = 0$.

$$x^2 - 4x - 5 = (x - 2)^2 - 9$$

$$(x - 2)^2 = 9, \text{ so } x - 2 = +3 \text{ or } -3$$

$$\text{Therefore } x = 5 \text{ or } x = -1.$$

Check both values in the original equation before moving on.

GUIDED PRACTICE

A. Complete the square

Follow the five-step recipe. Show the half and the square each time.

Question	Half of b	Square it	Completed-square form
1. $x^2 + 4x + 7$			
2. $x^2 + 10x + 3$			
3. $x^2 - 6x + 2$			
4. $x^2 - 12x + 5$			
5. $x^2 + 2x - 8$			
6. $x^2 - 2x - 3$			

CHECK BEFORE PAGE 9

Expand each answer. The x^2 term, x term and constant must all match the original. If the x term is wrong, check whether you halved b. If the constant is wrong, check whether you subtracted the square outside.

Confidence after this page: 1 2 3 4 5

If you chose 1 or 2, redo Questions 1 and 3 slowly before continuing.

GUIDED APPLICATION

B. Read information from the square

Complete the square first, then use (x - h) squared + k.

Quadratic	Completed-square form	Turning point	Minimum value	Line of symmetry
1. $y = x^2 + 6x + 2$				
2. $y = x^2 - 4x + 7$				
3. $y = x^2 + 8x + 10$				
4. $y = x^2 - 10x + 4$				

REMEMBER

The bracket sign looks opposite to the turning-point x-coordinate. $(x + 3)^2$ has x-coordinate -3; $(x - 5)^2$ has x-coordinate +5.

Explain one answer aloud

Choose one row. Explain why its minimum value is the outside constant and why its line of symmetry passes through the turning point.

My explanation:

INDEPENDENT PRACTICE

C. No sentence starters

Work without looking at the examples. Show every line.

1. Write $x^2 + 14x + 8$ in completed-square form.

2. Write $x^2 - 16x + 20$ in completed-square form.

3. Write $x^2 + 3x - 1$ in completed-square form.

4. For $y = x^2 + 2x - 6$, find the turning point and minimum value.

5. For $y = x^2 - 12x + 40$, state the line of symmetry and range.

6. Solve $x^2 + 8x + 7 = 0$ by completing the square.

7. Solve $x^2 - 6x + 5 = 0$ by completing the square.

8. Solve $x^2 + 4x - 2 = 0$. Give exact answers.

DO NOT OPEN THE ANSWERS YET

Put a star beside any question you are unsure about. Then attempt the exit check on the next page without notes.

EXIT CHECK - 10 MINUTES

Can you do it alone?

Close pages 2-10. This score decides what you should do next.

Question	Your answer
1. In $x^2 - 9x + 4$, state b.	
2. Expand $(x - 5)^2$.	
3. Complete the square: $x^2 + 12x + 1$.	
4. Complete the square: $x^2 - 2x + 6$.	
5. State the turning point of $y = (x + 4)^2 - 3$.	
6. State the minimum value and range in Question 5.	
7. Solve $(x - 1)^2 = 16$.	
8. Solve $x^2 + 10x + 9 = 0$ by completing the square.	

NOW UNLOCK PAGES 12-13

Mark in a different colour. Give one mark per exit question. For Question 6, both the minimum and range must be correct for the mark.

Score	What to do tomorrow
7-8	Ready for harder completed-square questions, including a coefficient before x squared.
5-6	Redo only the incorrect questions, then try a fresh four-question check.
0-4	Reread pages 3-7 and repeat Guided Practice A before moving on.

Guided and independent answers

Open only after completing pages 1-11. Correct the first wrong line.

A. Complete the square - page 8

Q	Half of b	Square	Answer
1	2	4	$(x + 2)^2 + 3$
2	5	25	$(x + 5)^2 - 22$
3	-3	9	$(x - 3)^2 - 7$
4	-6	36	$(x - 6)^2 - 31$
5	1	1	$(x + 1)^2 - 9$
6	-1	1	$(x - 1)^2 - 4$

B. Turning points - page 9

Q	Completed-square form	Turning point	Minimum	Symmetry
1	$(x + 3)^2 - 7$	(-3, -7)	-7	x = -3
2	$(x - 2)^2 + 3$	(2, 3)	3	x = 2
3	$(x + 4)^2 - 6$	(-4, -6)	-6	x = -4
4	$(x - 5)^2 - 21$	(5, -21)	-21	x = 5

C. Independent practice - page 10

- 1. $(x + 7)^2 - 41$
- 2. $(x - 8)^2 - 44$
- 3. $(x + 3/2)^2 - 13/4$
- 4. $(x + 1)^2 - 7$; turning point (-1, -7); minimum -7
- 5. $(x - 6)^2 + 4$; line x = 6; range y is greater than or equal to 4
- 6. $(x + 4)^2 = 9$; x = -1 or x = -7
- 7. $(x - 3)^2 = 4$; x = 5 or x = 1
- 8. $(x + 2)^2 = 6$; x = -2 + sqrt(6) or x = -2 - sqrt(6)

Exit-check answers and correction log

A wrong answer is a map: find the first step that went wrong.

Q	Answer	Review if wrong
1	$b = -9$	Page 2: the sign belongs to the coefficient.
2	$x^2 - 10x + 25$	Page 3: the middle term is twice -5.
3	$(x + 6)^2 - 35$	Page 4: halve 12, square 6, then compensate.
4	$(x - 1)^2 + 5$	Page 5: half of -2 is -1.
5	(-4, -3)	Page 6: bracket sign is opposite.
6	Minimum -3; range y is greater than or equal to -3	Page 6: a square cannot be negative.
7	$x = 5$ or $x = -3$	Page 7: take both square roots.
8	$(x + 5)^2 = 16$, so $x = -1$ or $x = -9$	Page 7: isolate the square, then use +4 and -4.

PASS RULE

7-8 means move on. 5-6 means correct and retry. 0-4 means repeat the core lesson. Do not hide a mistake: write the correction beside it in a different colour.

Correction log

Question	My first wrong line	Why it was wrong	Correct rule

Send for marking

Take clear photos of pages 8-11, including every line of working. Send them back with this sentence: **'Please mark my Maths P1 Quadratics Day 1 work and tell me my three priority corrections.'**

TOMORROW

If you pass, the next sheet should cover quadratics with a coefficient before x^2, discriminants and the number of real roots. If not, the next sheet should target only the mistakes recorded above.

AS MATHEMATICS - PURE 1 - DAY 2

Quadratics: level up

Coefficients before x squared and the discriminant

YOUR AIM TODAY

You will extend completing the square to expressions such as 2x squared + 8x + 1, then use the discriminant to decide how many real roots a quadratic has.

Block	What you do	Time
1	Retrieve yesterday's method	10 min
2	Learn the coefficient method	20 min
3	Learn and use the discriminant	20 min
4	Complete mixed independent practice	25 min
5	Exit check and self-mark	10 min

YOUR RULE

Keep page 9 closed until pages 1-8 are complete. Show every line. If Day 1 was difficult, use the short retrieval check before starting the new method.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: ___________________

RETRIEVAL

Can you still complete the square?

Attempt without notes. If stuck, revisit Day 1 pages 3-5.

Question	Your working and answer
1. Expand $(x + 4)^2$.	
2. Complete: $x^2 + 8x + 3$.	
3. Complete: $x^2 - 6x + 1$.	
4. State the turning point of $y = (x - 5)^2 - 2$.	
5. Solve $(x + 1)^2 = 9$.	

QUICK ANSWERS

1. $x^2 + 8x + 16$ 2. $(x + 4)^2 - 13$ 3. $(x - 3)^2 - 8$
4. $(5, -2)$ 5. $x = 2$ or $x = -4$

Score ___ / 5. If below 4, correct the errors before continuing.

NEW METHOD

When the x-squared coefficient is not 1

Factor the coefficient from the x-squared and x terms first.

Example 1: write $2x^2 + 8x + 1$ in completed-square form.

STEP 1 - FACTOR 2

$2x^2 + 8x + 1 = 2(x^2 + 4x) + 1$

Complete the square **inside** the bracket:

$$x^2 + 4x = (x + 2)^2 - 4$$

Now keep the outside 2:

$$2[(x + 2)^2 - 4] + 1 = 2(x + 2)^2 - 8 + 1$$

$$\text{Therefore: } 2x^2 + 8x + 1 = 2(x + 2)^2 - 7.$$

COMMON MISTAKE

The -4 is inside a bracket multiplied by 2, so it becomes -8. Do not write $2(x + 2)^2 - 4 + 1$.

Example 2

$$3x^2 - 12x + 5 = 3(x^2 - 4x) + 5$$

$$= 3[(x - 2)^2 - 4] + 5$$

$$= 3(x - 2)^2 - 7$$

Expand both final answers to check them.

GUIDED PRACTICE

A. Complete the square with a coefficient

Factor a first. Complete the square inside. Multiply back out.

Expression	Factor first	Complete inside	Final form
1. $2x^2 + 12x + 5$			
2. $3x^2 + 18x + 4$			
3. $2x^2 - 8x + 7$			
4. $4x^2 - 24x + 1$			
5. $5x^2 + 10x - 3$			

CHECK BY EXPANDING

Your final answer must return to the same x^2 coefficient, x coefficient and constant. If the constant is wrong, check whether the outside coefficient multiplied the compensation term.

NEW IDEA

The discriminant

Use it to decide the number of real roots without solving the equation.

For $ax^2 + bx + c = 0$, the discriminant is:

$$D = b^2 - 4ac$$

Value of D	Meaning	Graph
$D > 0$	Two distinct real roots	Crosses the x-axis twice
$D = 0$	One repeated real root	Touches the x-axis once
$D < 0$	No real roots	Does not meet the x-axis

Worked example

For $2x^2 + 3x - 2 = 0$: $a = 2$, $b = 3$, $c = -2$.

$$D = 3^2 - 4(2)(-2) = 9 + 16 = 25$$

Because $D > 0$, the equation has **two distinct real roots**.

SIGN WARNING

If c is negative, -4ac may become positive. Write brackets around negative values before calculating.

Say it aloud

Positive: two. Zero: one repeated. Negative: none real.

GUIDED PRACTICE

B. Use the discriminant

State a, b and c first. Then calculate D and interpret it.

Equation	a, b, c	D working	Number of real roots
1. $x^2 + 5x + 6 = 0$			
2. $x^2 - 6x + 9 = 0$			
3. $2x^2 + x + 4 = 0$			
4. $3x^2 - 7x + 2 = 0$			
5. $4x^2 + 4x + 1 = 0$			

INTERPRET, DO NOT STOP

A discriminant number is not the final response. Always finish with: two distinct real roots, one repeated real root, or no real roots.

INDEPENDENT PRACTICE

C. Mixed quadratics

Choose the method required by the command word.

1. Write $2x^2 + 16x + 3$ in completed-square form.

2. Write $3x^2 - 30x + 8$ in completed-square form.

3. Hence state the turning point and minimum value of $y = 3x^2 - 30x + 8$.

4. Find the discriminant of $x^2 + 2x + 5 = 0$ and state the number of real roots.

5. Find the values of k for which $x^2 + 4x + k = 0$ has a repeated root.

6. The equation $2x^2 + px + 8 = 0$ has no real roots. Write an inequality involving p.

KEEP THE ANSWER PAGE CLOSED

Attempt every question. Put a star beside uncertainty, then complete the exit check.

EXIT CHECK

Day 2 checkpoint

No notes. One mark per row; Question 6 is worth two marks.

Question	Answer
1. Complete: $2x^2 + 8x - 1$.	
2. State the turning point of your answer to Question 1.	
3. Calculate D for $x^2 - 4x + 4 = 0$.	
4. How many real roots does Question 3 have?	
5. Calculate D for $3x^2 + 2x + 5 = 0$ and interpret it.	
6. Find k if $x^2 + 6x + k = 0$ has a repeated root. Show why.	

UNLOCK PAGE 9

Mark in a different colour. 6-7: ready to progress. 4-5: correct and retry. 0-3: repeat pages 3-6.

ANSWER SECTION

Answers, corrections and next step

Correct the first wrong line, not only the final answer.

A. Coefficient practice

- 1. $2(x + 3)^2 - 13$
- 2. $3(x + 3)^2 - 23$
- 3. $2(x - 2)^2 - 1$
- 4. $4(x - 3)^2 - 35$
- 5. $5(x + 1)^2 - 8$

B. Discriminant practice

Q	D	Conclusion
1	1	Two distinct real roots
2	0	One repeated real root
3	-31	No real roots
4	25	Two distinct real roots
5	0	One repeated real root

C. Independent practice

- 1. $2(x + 4)^2 - 29$
- 2. $3(x - 5)^2 - 67$
- 3. Turning point $(5, -67)$; minimum -67.
- 4. $D = -16$, so no real roots.
- 5. $D = 16 - 4k = 0$, so $k = 4$.
- 6. $D = p^2 - 64 < 0$, so $p^2 < 64$, equivalently $-8 < p < 8$.

Exit check

- 1. $2(x + 2)^2 - 9$
- 2. $(-2, -9)$
- 3. $D = 0$
- 4. One repeated real root
- 5. $D = -56$, so no real roots
- 6. $D = 36 - 4k = 0$, so $k = 9$

SEND FOR MARKING

Send clear photos of pages 4, 6, 7 and 8 with every line of working. Ask: 'Please mark my Maths P1 Day 2 work and identify my three priority corrections.'

AS MATHEMATICS - PURE 1 - DAY 3

See the whole quadratic

Graphs, inequalities and discriminant parameters

YOUR AIM TODAY

You will connect factors, roots, turning points and graphs; solve quadratic inequalities; and use the discriminant to find conditions on a parameter.

Block	What you do	Time
1	Day 2 recovery check	10 min
2	Connect equations to graphs	20 min
3	Learn quadratic inequalities	20 min
4	Learn parameter conditions	15 min
5	Independent practice and exit check	25 min

YOUR RULE

Keep pages 10-11 closed until pages 1-9 are complete. Draw a small sketch for every inequality. Show the discriminant inequality before solving for a parameter.

RECOVERY OPTION

If you score below 4/5 on page 2, spend ten minutes correcting it before starting the new content.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

RECOVERY CHECK

Day 2 foundations

No notes for the first attempt.

Question	Working and answer
1. Complete: $2x^2 + 12x + 1$.	
2. State its turning point and minimum value.	
3. Find D for $x^2 + 3x + 5 = 0$.	
4. State the number of real roots in Question 3.	
5. Find k if $x^2 + 8x + k = 0$ has a repeated root.	

QUICK ANSWERS

1. $2(x + 3)^2 - 17$ 2. $(-3, -17)$, minimum -17
3. $D = -11$ 4. No real roots 5. $64 - 4k = 0$, so $k = 16$

Score ___ / 5. Correct every error before page 3.

CONNECTION 1

One quadratic, three useful forms

Choose the form that reveals the information you need.

Form	Example	What it reveals
Expanded	$y = x^2 - 6x + 5$	y-intercept c = 5; coefficients for D
Factorised	$y = (x - 1)(x - 5)$	Roots x = 1 and x = 5
Completed square	$y = (x - 3)^2 - 4$	Turning point (3, -4); axis x = 3

Why the graph crosses at the roots

On the x-axis, y = 0. Therefore the x-intercepts solve $x^2 - 6x + 5 = 0$. From $(x - 1)(x - 5) = 0$, the roots are 1 and 5.

Feature	Value	Reason
x-intercepts	(1, 0) and (5, 0)	Set y = 0
y-intercept	(0, 5)	Set x = 0
turning point	(3, -4)	Read completed-square form
axis of symmetry	x = 3	Vertical line through turning point
shape	Opens upward	Coefficient of x^2 is positive

SYMMETRY CHECK

The axis lies halfway between the roots: (1 + 5) / 2 = 3.

Sketch this graph on paper and label all four intercept/turning-point coordinates.

CONNECTION 2

How roots and the discriminant shape the graph

The sign of D tells you how the graph meets the x-axis.

D	Roots	Graph behaviour	Example
$D > 0$	Two distinct real roots	Crosses the x-axis twice	$y = x^2 - 5x + 6$
$D = 0$	One repeated root	Touches the x-axis at the turning point	$y = x^2 - 4x + 4$
$D < 0$	No real roots	Does not meet the x-axis	$y = x^2 + 2x + 5$

Worked comparison

- $x^2 - 5x + 6 = (x - 2)(x - 3)$: roots 2 and 3, so two crossings.
- $x^2 - 4x + 4 = (x - 2)^2$: repeated root 2, so one touch.
- $x^2 + 2x + 5 = (x + 1)^2 + 4$: minimum 4, so it stays above the axis.

POWERFUL CONNECTION

Completed-square form explains *where* the turning point is. The discriminant explains *how many* x-axis meetings occur. Factorised form gives the roots directly when factorisation is possible.

Quick check

Quadratic	D sign	Number of x-axis meetings
1. $x^2 + 4x + 4$		
2. $x^2 + 4x + 8$		
3. $x^2 - x - 6$		

NEW METHOD

Solve a quadratic inequality

Find the roots, sketch the sign, then select the required region.

Example 1: solve $x^2 - 5x + 6 > 0$.

Factorise: $(x - 2)(x - 3) > 0$, so the boundary roots are 2 and 3.

Interval	Test x	Value of (x - 2)(x - 3)	Sign
$x < 2$	0	$(-2)(-3) = 6$	+
$2 < x < 3$	2.5	$(0.5)(-0.5) = -0.25$	-
$x > 3$	4	$(2)(1) = 2$	+

We need values where the quadratic is positive, so:

$$x < 2 \text{ or } x > 3$$

OUTSIDE FOR POSITIVE

For an upward-opening quadratic with two roots, the graph is above the x-axis outside the roots and below it between the roots.

Boundary rule

- Use open boundaries for < or >. The roots are not included.
- Use closed boundaries for <= or >=. The roots are included because the expression can equal 0.

NEVER DIVIDE BY AN EXPRESSION CONTAINING X

Its sign may be unknown, so dividing could reverse the inequality incorrectly. Factorise and use intervals instead.

WORKED EXAMPLES

Inside, outside and downward-opening

Use a sketch or sign table every time.

Example 2: solve $x^2 - x - 6 <= 0$.

$(x - 3)(x + 2) <= 0$. Roots: -2 and 3. The upward graph is on or below the axis between the roots.

$$-2 <= x <= 3$$

Example 3: solve $-x^2 + 5x - 6 > 0$.

Multiply by -1 and reverse the inequality:

$$x^2 - 5x + 6 < 0$$

$$(x - 2)(x - 3) < 0, \text{ so } 2 < x < 3.$$

WHY REVERSE?

Multiplying or dividing both sides of an inequality by a negative number reverses < and >.

Mini-practice

Question	Roots	Solution
1. $x^2 - 7x + 12 > 0$		
2. $x^2 + x - 6 < 0$		
3. $-x^2 + 4x + 5 >= 0$		
4. $x^2 - 10x + 25 <= 0$		

PARAMETER SKILL

Turn root conditions into inequalities

Translate the words into a condition on D.

Words	Discriminant condition
Two distinct real roots	$D > 0$
One repeated real root	$D = 0$
At least one real root	$D >= 0$
No real roots	$D < 0$

Example 4

Find the values of k for which $x^2 + 2x + k = 0$ has no real roots.

$$a = 1, b = 2, c = k$$

$$D = 2^2 - 4(1)(k) = 4 - 4k$$

No real roots means $D < 0$:

$$4 - 4k < 0$$

$$-4k < -4, \text{ so } k > 1$$

SIGN REVERSAL AGAIN

Dividing $-4k < -4$ by -4 reverses the sign, giving $k > 1$.

Example 5

For two distinct real roots of $x^2 + kx + 9 = 0$:

$$D = k^2 - 36 > 0$$

$$(k - 6)(k + 6) > 0, \text{ so } k < -6 \text{ or } k > 6.$$

GUIDED PRACTICE

A. Graphs, inequalities and parameters

Write the reason beside every conclusion.

Question	Working and answer
1. For $y = x^2 - 8x + 12$, find the roots, turning point, y-intercept and axis of symmetry.	
2. Solve $x^2 - 8x + 12 < 0$.	
3. Solve $x^2 + 2x - 15 >= 0$.	
4. Find k if $x^2 - 4x + k = 0$ has a repeated root.	
5. Find the values of k for which $x^2 + 6x + k = 0$ has no real roots.	
6. Find the values of p for which $x^2 + px + 4 = 0$ has two distinct real roots.	

METHOD CHECK

Questions 2-3 need roots plus an interval sketch. Questions 4-6 need D = 0, D < 0 or D > 0 before solving the parameter condition.

INDEPENDENT CHECKPOINT

B. Do it without prompts

Do not open the answers until every question has an attempt.

1. Sketch $y = x^2 - 2x - 8$, labelling both roots, the turning point and the y-intercept.

2. Solve $x^2 - 2x - 8 > 0$.

3. Solve $2x^2 - 10x + 12 <= 0$.

4. Find the values of k for which $x^2 + 4x + k = 0$ has two distinct real roots.

5. Find the values of m for which $2x^2 + mx + 3 = 0$ has no real roots.

6. Explain in one sentence why $y = 3(x - 2)^2 + 5$ has no real roots.

UNLOCK PAGES 10-11

Mark in a different colour. Then complete the correction log and send the requested pages for marking.

Guided answers

Compare the method, not only the last line.

Page 4 quick check

Q	D	x-axis meetings
1	0	One touch
2	-16	None
3	25	Two crossings

Page 6 mini-practice

- 1. Roots 3 and 4; solution $x < 3$ or $x > 4$.
- 2. Roots -3 and 2; solution $-3 < x < 2$.
- 3. Rewrite $x^2 - 4x - 5 <= 0$; roots -1 and 5; solution $-1 <= x <= 5$.
- 4. $(x - 5)^2 <= 0$ only when $x = 5$.

Page 8 guided practice

Q	Answer
1	$(x - 2)(x - 6)$, so roots 2 and 6; $(x - 4)^2 - 4$, so turning point (4, -4); y-intercept (0, 12); axis $x = 4$.
2	$2 < x < 6$.
3	Roots -5 and 3; solution $x <= -5$ or $x >= 3$.
4	$D = 16 - 4k = 0$, so $k = 4$.
5	$D = 36 - 4k < 0$, so $k > 9$.
6	$D = p^2 - 16 > 0$, so $p < -4$ or $p > 4$.

COMMON CORRECTIONS

For inequalities, check whether endpoints are included. For parameter questions, write the correct D condition before manipulating the inequality. For graphs, label coordinates rather than only drawing a curve.

Independent answers and correction log

Find the first wrong line and repair the rule.

Q	Answer
1	$y = (x - 4)(x + 2)$: roots $(-2, 0)$ and $(4, 0)$; $y = (x - 1)^2 - 9$: turning point $(1, -9)$; y-intercept $(0, -8)$. Upward-opening graph.
2	$x < -2$ or $x > 4$.
3	$2(x - 2)(x - 3) <= 0$, so $2 <= x <= 3$.
4	$D = 16 - 4k > 0$, so $k < 4$.
5	$D = m^2 - 24 < 0$, so $m^2 < 24$; therefore $-2\text{sqrt}(6) < m < 2\text{sqrt}(6)$.
6	Because $3(x - 2)^2 + 5$ is always at least 5, the graph never reaches $y = 0$.

Correction log

Question	My first wrong line	Correct rule

PASS RULE

5-6 correct on page 9: ready to move beyond quadratics. 3-4: correct and retry tomorrow. 0-2: repeat pages 5-8 before new content.

SEND FOR MARKING

Send clear photos of pages 8, 9 and 11, plus the graph from Question 1. Ask: 'Please mark my Maths P1 Day 3 work and identify my three priority corrections.'

AS MATHEMATICS - PURE 1 - DAY 4

Functions as machines

Notation, composites, inverses, domain and range

YOUR AIM TODAY

You will read function notation, substitute accurately, combine two functions and reverse a one-to-one function.

YOUR RULE

Keep page 8 closed until pages 1-7 are complete. Work from the inside function outward. For an inverse, swap x and y, then rearrange.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

RETRIEVAL

Quadratics checkpoint

Short review before a new topic.

Question	Answer
1. Complete: $x^2 - 6x + 2$.	
2. Solve $x^2 - 5x + 6 > 0$.	
3. Find k if $x^2 + 4x + k = 0$ has a repeated root.	
4. State the turning point of $y = (x + 2)^2 - 7$.	

CHECK

1. $(x - 3)^2 - 7$ 2. $x < 2$ or $x > 3$ 3. $k = 4$ 4. $(-2, -7)$

FOUNDATION

Function notation and substitution

A function gives exactly one output for each allowed input.

MACHINE IDEA

If $f(x) = 2x + 3$, the machine doubles the input and adds 3. Thus $f(4) = 2(4) + 3 = 11$.

Notation	Meaning	Example
$f(x)$	Output of f when input is x	$f(x)=2x+3$
$f(5)$	Replace every x by 5	$f(5)=13$
$f(a)$	Replace every x by a	$f(a)=2a+3$
$f(x+1)$	Replace every x by $(x+1)$	$2(x+1)+3=2x+5$

BRACKET WARNING

For $f(x)=x^2-4x$, $f(-2)=(-2)^2-4(-2)=12$. Use brackets around negative inputs.

Domain and range

- Domain: the allowed input values.
- Range: the output values actually produced.
- For $f(x)=1/(x-3)$, $x=3$ is excluded from the domain because division by zero is impossible.

Question	Your answer
1. $f(x)=3x-5$. Find $f(7)$.	
2. $g(x)=x^2+2$. Find $g(-3)$.	
3. Find $f(a+2)$.	
4. State the excluded value for $h(x)=1/(x+4)$.	

One machine feeds into another

fg(x) means f(g(x)): perform g first.

Let $f(x)=2x+1$ and $g(x)=x^2-3$.

$$fg(2)=f(g(2))$$

$g(2)=1$, then $f(1)=3$, so $fg(2)=3$.

Algebraic composite

$$fg(x)=f(x^2-3)=2(x^2-3)+1=2x^2-5$$

$$gf(x)=g(2x+1)=(2x+1)^2-3=4x^2+4x-2$$

ORDER MATTERS

Usually fg(x) is not equal to gf(x). Read from the function nearest x: g acts first in f(g(x)).

Question	Working and answer
1. Find fg(3).	
2. Find gf(3).	
3. Find fg(x).	
4. Find gf(x).	

INVERSE FUNCTIONS

Undo the original function

An inverse reverses every operation in the opposite order.

For $f(x)=3x-5$:

$$y=3x-5$$

Swap x and y: $x=3y-5$

Rearrange: $y=(x+5)/3$

Therefore $f^{-1}(x)=(x+5)/3$.

CHECK BY COMPOSITION

$f^{-1}(f(x))=[(3x-5)+5]/3=x$. Getting x proves the functions undo each other.

NOT A RECIPROCAL

$f^{-1}(x)$ does not mean $1/f(x)$. It means inverse function.

Function	Find the inverse
1. $f(x)=2x+7$	
2. $g(x)=5-3x$	
3. $h(x)=(x-4)/2$	
4. $p(x)=x^2$, $x>=0$	
5. $q(x)=(x+1)^2$, $x>=-1$	

GUIDED PRACTICE

A. Functions and their inverses

Show substitution brackets and composite order.

Question	Working and answer
1. $f(x)=4x-1$. Find $f(-2)$ and $f(a+1)$.	
2. $g(x)=x^2+1$. Find $g(3)$ and $g(-3)$.	
3. With f and g above, find $fg(2)$ and $gf(2)$.	
4. Find $f^{-1}(x)$.	
5. Show that $f^{-1}(f(x))=x$.	
6. State the range of $g(x)=x^2+1$ for real x.	

SELF-CHECK

Equal outputs $g(3)=g(-3)$ show that g is not one-to-one on all real numbers. A restricted domain is needed before it has an inverse.

INDEPENDENT CHECKPOINT

B. No prompts

Attempt all six before unlocking the answers.

1. $f(x)=2x+5$ and $g(x)=3-x$. Find $fg(x)$ and $gf(x)$.

2. Solve $fg(x)=9$.

3. Find f inverse and verify f inverse of $f(x)=x$.

4. $h(x)=(x-1)$ squared, $x>=1$. Find h inverse.

5. State the domain of $r(x)=1/(2x-6)$.

6. For $p(x)=x$ squared-4, $x>=0$, state the range and find p inverse.

UNLOCK PAGE 8

Mark in another colour and complete the correction log.

ANSWER SECTION

Answers and correction log

Repair the first wrong step.

Page 3

- 1. 16. 2. 11. 3. 3a+1. 4. x=-4 is excluded.

Page 4

- 1. fg(3)=f(6)=13. 2. gf(3)=g(7)=46. 3. 2x squared-5. 4. 4x squared+4x-2.

Page 5

- 1. (x-7)/2. 2. (5-x)/3. 3. 2x+4. 4. sqrt(x), x>=0. 5. sqrt(x)-1, x>=0.

Page 6

- 1. f(-2)=-9; f(a+1)=4a+3. 2. Both outputs are 10. 3. fg(2)=19; gf(2)=50. 4. f inverse=(x+1)/4. 5. [(4x-1)+1]/4=x. 6. range g>=1.

Page 7

Q	Answer
1	fg(x)=11-2x; gf(x)=-2x-2
2	11-2x=9, so x=1
3	f inverse=(x-5)/2; f inverse(f(x))=x
4	h inverse(x)=1+sqrt(x), x>=0
5	x is not equal to 3
6	Range p>=-4; p inverse(x)=sqrt(x+4), x>=-4

Question	First wrong step	Correct rule

SEND FOR MARKING

Send pages 6, 7 and 8. Ask for your three priority corrections.

AS MATHEMATICS - PURE 1 - DAY 5

Coordinate geometry

Gradients, distances, midpoints and straight lines

YOUR AIM TODAY

Build straight-line geometry from the coordinate basics.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

The ideas you need

A point is written (x, y). A line is controlled by its gradient and one point.

- Gradient $m=(\text{change in } y)/(\text{change in } x)$.
- Midpoint of (x_1,y_1) and (x_2,y_2) is $((x_1+x_2)/2,(y_1+y_2)/2)$.
- Distance is $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$.
- Line through (x_1,y_1): $y-y_1=m(x-x_1)$.
- Parallel lines have equal gradients; perpendicular gradients multiply to -1.

MEMORY RULE

Find the gradient first, then substitute one known point.

Line through two points

WORKED STEPS

Through (2,3) and (6,11): $m=(11-3)/(6-2)=2$. So $y-3=2(x-2)$, hence $y=2x-1$.

Perpendicular line

WORKED STEPS

A line perpendicular to $y=3x+4$ has gradient $-1/3$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Find the gradient from (1,2) to (5,10).	
2. Find the midpoint of (-2,5) and (6,-1).	
3. Find the distance between (1,1) and (4,5).	
4. Find the equation through (3,2) with gradient 4.	
5. Find the equation through (1,5) and (3,1).	
6. Find the perpendicular bisector of (0,0) and (4,2).	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	m=(10-2)/(5-1)=2.
2	(2,2).
3	sqrt(3^2+4^2)=5.
4	y-2=4(x-3), so y=4x-10.
5	m=-2; y-5=-2(x-1), so y=-2x+7.
6	Midpoint (2,1); segment gradient 1/2, so perpendicular gradient -2. y-1=-2(x-2), so y=-2x+5.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.

AS MATHEMATICS - PURE 1 - DAY 6

Circles

Centre, radius, equations and tangents

YOUR AIM TODAY

Read and build circle equations, then use radius-tangent geometry.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

The ideas you need

A circle is every point the same distance from its centre.

- Centre (a,b), radius r: $(x-a)^2+(y-b)^2=r^2$.
- The signs reverse inside brackets: $(x+3)^2$ gives centre $x=-3$.
- Expand only when asked; completed-square form shows the geometry.
- A tangent is perpendicular to the radius at the point of contact.

MEMORY RULE

Centre signs reverse; tangent gradient is the negative reciprocal of radius gradient.

Read a circle

WORKED STEPS

$(x-2)^2+(y+1)^2=25$ has centre (2,-1) and radius 5.

Tangent

WORKED STEPS

Centre (0,0), point (3,4): radius gradient 4/3, tangent gradient -3/4; $y-4=(-3/4)(x-3)$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. State centre and radius of (x+1)^2+(y-4)^2=9.	
2. Write the circle with centre (3,-2), radius 4.	
3. Does (4,3) lie on x^2+y^2=25?	
4. Complete squares: x^2+y^2-6x+4y-12=0.	
5. Find the tangent to x^2+y^2=25 at (3,4).	
6. Find intersections of x^2+y^2=25 with y=4.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	Centre (-1,4), radius 3.
2	$(x-3)^2+(y+2)^2=16$.
3	Yes: 16+9=25.
4	$(x-3)^2+(y+2)^2=25$.
5	$y-4=(-3/4)(x-3)$, or $3x+4y=25$.
6	$x^2+16=25$, so $x=3$ or -3: (3,4), (-3,4).

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.

AS MATHEMATICS - PURE 1 - DAY 7

Radians

Angles, arcs, sectors and segments

YOUR AIM TODAY

Use radians confidently in exact geometry problems.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

The ideas you need

Radians measure an angle using arc length divided by radius.

- pi radians=180 degrees.
- Degrees to radians: multiply by pi/180.
- Arc length s=r theta.
- Sector area=(1/2)r^2 theta.
- Segment area=sector area-triangle area=(1/2)r^2(theta-sin theta).

MEMORY RULE

The arc and sector formulas require theta in radians.

Convert

WORKED STEPS

60 degrees=60(pi/180)=pi/3.

Sector

WORKED STEPS

r=6, theta=1.2: arc=7.2; area=0.5(36)(1.2)=21.6.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Convert 135 degrees to radians.	
2. Convert 5pi/6 radians to degrees.	
3. Find arc length when r=8, theta=0.75.	
4. Find sector area when r=5, theta=1.4.	
5. Find the perimeter of that sector.	
6. Find segment area for r=4, theta=pi/3.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	3pi/4.
2	150 degrees.
3	s=8(0.75)=6.
4	0.5(25)(1.4)=17.5.
5	2r+s=10+7=17.
6	0.5(16)(pi/3-sin(pi/3))=8pi/3-4sqrt(3).

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.

AS MATHEMATICS - PURE 1 - DAY 8

Trigonometry

Identities and equations

YOUR AIM TODAY

Simplify expressions and solve trigonometric equations over an interval.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

The ideas you need

An identity is true for every allowed angle; an equation is true only for selected angles.

- $\sin^2 x+\cos^2 x=1$.
- $\tan x=\sin x/\cos x$.
- For $\sin x=k$, use the sine symmetry; for $\cos x=k$, use cosine symmetry.
- For $\tan x=k$, solutions repeat every 180 degrees.
- Always check the stated interval.

MEMORY RULE

Find the reference angle, choose the correct quadrants, then check the interval.

Identity

WORKED STEPS

$(1-\cos^2 x)/\sin x=\sin^2 x/\sin x=\sin x$.

Equation

WORKED STEPS

$2\sin x=1$ on 0 to 360: $\sin x=1/2$, so $x=30,150$ degrees.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Simplify sin x/cos x.	
2. Simplify 1-sin^2 x.	
3. Solve sin x=sqrt(3)/2 for 0<=x<=360 degrees.	
4. Solve cos x=-1/2 on the same interval.	
5. Solve tan x=1 on the same interval.	
6. Solve 2cos^2 x-1=0 on the same interval.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	tan x.
2	cos^2 x.
3	60, 120 degrees.
4	120, 240 degrees.
5	45, 225 degrees.
6	cos^2 x=1/2, so x=45,135,225,315 degrees.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.

AS MATHEMATICS - PURE 1 - DAY 9

Binomial expansion

Coefficients and selected terms

YOUR AIM TODAY

Expand positive integer powers and locate a required coefficient.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

The ideas you need

The coefficients in $(a+b)^n$ come from combinations.

- General term: $C(n,r)a^{(n-r)}b^r$.
- r starts at 0, so the $(r+1)$th term uses r.
- Signs alternate for $(a-b)^n$.
- For a selected coefficient, track both number and power of x.

MEMORY RULE

Write $C(n,r)$, then the first term power falls while the second rises.

Expansion

WORKED STEPS

$(1+2x)^3=1+3(2x)+3(2x)^2+(2x)^3=1+6x+12x^2+8x^3$.

Selected term

WORKED STEPS

x^2 in $(2+x)^5$ uses $r=2$: $C(5,2)2^3x^2=80x^2$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Expand $(1+x)^4$.	
2. Expand $(2-x)^3$.	
3. Find coefficient of x^2 in $(1+3x)^5$.	
4. Find coefficient of x^3 in $(2+x)^6$.	
5. Find first four terms of $(1-2x)^5$.	
6. Find constant term in $(x+2/x)^4$.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$1+4x+6x^2+4x^3+x^4$.
2	$8-12x+6x^2-x^3$.
3	$C(5,2)3^2=90$.
4	$C(6,3)2^3=160$.
5	$1-10x+40x^2-80x^3$.
6	Power $x^{(4-r)}x^{(-r)}=x^{(4-2r)}$; constant when $r=2$. $C(4,2)2^2=24$.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.

AS MATHEMATICS - PURE 1 - DAY 10

Sequences and series

Arithmetic and geometric progressions

YOUR AIM TODAY

Find terms and sums, then decide which model applies.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

The ideas you need

A progression follows a fixed rule from one term to the next.

- AP: common difference d; nth term a+(n-1)d.
- AP sum: n/2[2a+(n-1)d].
- GP: common ratio r; nth term ar^(n-1).
- GP sum: a(1-r^n)/(1-r).
- Infinite GP sum a/(1-r) only when |r|<1.

MEMORY RULE

Subtract neighbouring terms for an AP; divide them for a GP.

AP

WORKED STEPS

3,7,11,... has a=3,d=4. Term 20=3+19(4)=79.

GP

WORKED STEPS

5,10,20,... has a=5,r=2. Sum first 4=5(1-16)/(1-2)=75.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Find term 15 of 4,7,10,...	
2. Find sum first 20 terms of 2,6,10,...	
3. Find term 8 of 3,6,12,...	
4. Find sum first 6 terms of 2,6,18,...	
5. Find infinite sum 12+6+3+...	
6. In an AP, term 3=11 and term 8=31. Find a and d.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	a=4,d=3: 4+14(3)=46.
2	n/2[2a+(n-1)d]=10[4+76]=800.
3	3(2^7)=384.
4	2(1-3^6)/(1-3)=728.
5	12/(1-1/2)=24.
6	a+2d=11, a+7d=31, so d=4 and a=3.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.

AS MATHEMATICS - PURE 1 - DAY 11

Differentiation

Gradients, tangents and normals

YOUR AIM TODAY

Differentiate powers and use the derivative as a gradient.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

The ideas you need

The derivative tells you how fast y changes as x changes.

- $d/dx(x^n)=n x^{(n-1)}$.
- Constants differentiate to 0.
- Differentiate every term separately.
- At x=a, substitute a into dy/dx for the tangent gradient.
- Normal gradient is the negative reciprocal of tangent gradient.

MEMORY RULE

Bring the power down, then reduce it by one.

Derivative

WORKED STEPS

$y=3x^4-2x^2+7$ gives $dy/dx=12x^3-4x$.

Tangent

WORKED STEPS

$y=x^2+1$ at $x=2$: point (2,5), gradient 4, so $y-5=4(x-2)$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Differentiate x^5.	
2. Differentiate $4x^3-3x+8$.	
3. Find gradient of $y=x^3-2x$ at $x=2$.	
4. Find tangent to $y=x^2+3$ at $x=1$.	
5. Find normal to $y=x^2$ at $x=2$.	
6. If $y=2x^3+kx$ has gradient 13 at $x=1$, find k.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	5x^4.
2	12x^2-3.
3	dy/dx=3x^2-2; gradient 10.
4	Point (1,4), m=2: y=2x+2.
5	Tangent m=4, normal m=-1/4; point (2,4): y-4=-(1/4)(x-2).
6	dy/dx=6x^2+k; 6+k=13, so k=7.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.

AS MATHEMATICS - PURE 1 - DAY 12

Stationary points

Turning points and optimisation

YOUR AIM TODAY

Find, classify and interpret stationary points.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

The ideas you need

At a stationary point, the tangent is horizontal, so $dy/dx=0$.

- Solve $dy/dx=0$ to find stationary x-values.
- Substitute into y to find coordinates.
- Second derivative positive: local minimum; negative: local maximum.
- A sign change + to - is a maximum; - to + is a minimum.

MEMORY RULE

Differentiate, equal zero, solve, substitute, classify.

Classify

WORKED STEPS

$y=x^2-4x+1$. $dy/dx=2x-4=0$ gives $x=2$; $y=-3$. $d^2y/dx^2=2>0$, so minimum $(2,-3)$.

Cubic

WORKED STEPS

$y=x^3-3x$: $dy/dx=3x^2-3=0$ gives $x=\pm 1$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Find stationary point of $y=x^2+6x+2$.	
2. Classify it.	
3. Find stationary x-values of $y=x^3-12x$.	
4. Find and classify both stationary points in Q3.	
5. Find minimum value of $x^2-8x+20$.	
6. Rectangle perimeter is 20. If sides are x and $10-x$, find maximum area.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	dy/dx=2x+6=0, x=-3; y=-7.
2	d2y/dx2=2>0, so a minimum.
3	3x^2-12=0, so x=+-2.
4	At x=-2, y=16 and second derivative -12: maximum. At x=2, y=-16 and second derivative 12: minimum.
5	Stationary x=4; minimum value 4.
6	A=x(10-x)=10x-x^2. A'=10-2x=0 gives x=5; maximum area 25.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.

AS MATHEMATICS - PURE 1 - DAY 13

Integration

Antiderivatives, constants and areas

YOUR AIM TODAY

Reverse differentiation and evaluate definite integrals.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

The ideas you need

Integration rebuilds a function from its rate of change.

- Integral $x^n dx = x^{(n+1)}/(n+1)+C$ for n not equal to -1.
- Add C for an indefinite integral.
- Use a given point to find C.
- For a definite integral, evaluate upper minus lower.
- Area may require splitting where the curve crosses the axis.

MEMORY RULE

Increase the power by one, divide by the new power.

Indefinite

WORKED STEPS

Integral $(6x^2-4) dx=2x^3-4x+C$.

Definite

WORKED STEPS

Integral from 0 to 2 of $3x^2 dx=[x^3]0$ to $2=8$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Integrate 4x^3.	
2. Integrate 6x^2-2x+5.	
3. Given dy/dx=2x+3 and y=5 at x=1, find y.	
4. Evaluate integral 0 to 3 of 2x dx.	
5. Evaluate integral 1 to 2 of (3x^2+1) dx.	
6. Find area under y=4-x^2 from x=0 to x=2.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	x^4+C.
2	2x^3-x^2+5x+C.
3	y=x^2+3x+C; 5=1+3+C, so C=1. y=x^2+3x+1.
4	[x^2]0 to 3=9.
5	[x^3+x]1 to 2=(10)-(2)=8.
6	[4x-x^3/3]0 to 2=8-8/3=16/3.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.

AS MATHEMATICS - PURE 1 - DAY 14

P1 consolidation

A mixed checkpoint from quadratics to calculus

YOUR AIM TODAY

Retrieve methods without topic labels and diagnose the next weak area.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

The ideas you need

Mixed practice trains the hardest exam skill: selecting the method yourself.

- Read the command word and write the topic beside each question only after attempting it.
- If stuck, write the relevant formula before doing arithmetic.
- Check algebra, interval, units and exact form.
- Target: at least 5 of 6 fully correct.

MEMORY RULE

Choose method, show structure, calculate, check.

Model selection

WORKED STEPS

A tangent question needs differentiation; an arc needs radians; a repeated root needs discriminant zero.

Repair rule

WORKED STEPS

Do not erase an error. Mark the first wrong line, state the rule, then redo the whole question cleanly.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Solve x^2-7x+10=0.	
2. Find inverse of f(x)=3x-4.	
3. Find line through (2,1) perpendicular to y=2x+3.	
4. Find coefficient of x^2 in (1+2x)^4.	
5. Find stationary point of y=x^2-6x+4.	
6. Evaluate integral 0 to 2 of (3x^2+2) dx.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	(x-5)(x-2)=0, so x=2 or 5.
2	f^-1(x)=(x+4)/3.
3	Gradient -1/2; y-1=-(1/2)(x-2), so y=-x/2+2.
4	C(4,2)2^2=24.
5	dy/dx=2x-6=0 gives x=3; y=-5. Minimum.
6	[x^3+2x]0 to 2=8+4=12.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.